

AS.110.106 Calculus I (Biology and Social Sciences)

Full Course Design Package

Johns Hopkins University, Department of Mathematics
Redesigned for a 12-week lecture-format term

September 9, 2026

Abstract

This document reverse-engineers the AS.110.106 Calculus I (Bio. & Soc. Sci.) syllabus, based on *Calculus for Biology and Medicine*, 4th ed. (Neuhauser & Roper), into a complete 12-week course design: module structure, topic-level learning objectives, a prerequisite dependency map, a difficulty analysis with mitigations, an assessment schedule, and worked example questions.

Assumptions

- The syllabus gives week ranges as “2– weeks,” “2 weeks,” “3– weeks,” “2+ weeks,” “2 weeks,” “1+ week.” The minimum values sum to exactly 12 weeks, so this design uses the low end of each range to fit the stated 12-week term.
- The syllabus does not specify exam dates or a grading breakdown. This design assumes a standard two-midterm-plus-final structure, since none is stated.
- The syllabus does not specify contact hours per week; this design assumes a standard lecture cadence (roughly 2–3 sessions per week) without altering topic order.
- Chapter 1 is treated as an assumed-background review (functions, domain/range, composition, inverses, transformations, symmetry) since the syllabus lists it only as “Chapter 1” with no subsections.

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Course Structure Overview

Table 1: Twelve-week module map

Weeks	Module	Goal	Topics
1–2	A: Review of Functions & Growth Models	Refresh algebraic/functional foundations and introduce exponential and sequence-based models needed for later biological applications.	Ch. 1; 2.1; 2.2

Weeks	Module	Goal	Topics
3–4	B: Limits and Continuity	Develop rigorous and intuitive understanding of limiting behavior and continuity as the foundation of the derivative.	3.1–3.5
5–7	C: Derivatives	Build differentiation from first principles to fluent symbolic computation across all elementary function types.	4.1–4.11
8–9	D: Applications of Differentiation	Use derivatives to analyze function behavior and solve optimization problems.	5.1–5.5
10–11	E: Integration	Introduce the definite integral and connect differentiation and integration via the FTC.	5.10; 6.1–6.3
12	F: Applications of the Integral	Extend the integration toolkit with substitution, parts, and partial fractions for realistic models.	7.1–7.3

1 Module A: Review of Functions and Growth Models (Weeks 1–2)

1.1 Chapter 1: Review of Functions

Core Concepts

- Definition of a function; domain and range
- Composition of functions and inverse functions
- Polynomial, rational, trigonometric, exponential, and logarithmic function families
- Piecewise-defined functions
- Transformations: shifts, stretches, reflections
- Even/odd symmetry

Learning Objectives (students will be able to)

1. Identify the domain and range of algebraic, trigonometric, and piecewise functions.
2. Compute compositions $f \circ g$ and determine when a function is invertible.
3. Sketch transformations of a parent function given shift and scale parameters.
4. Classify a function as even, odd, or neither.

Example Questions

1. (*Recall*) State the domain of $f(x) = \frac{1}{\sqrt{x-3}}$.

Answer: $x > 3$, i.e. $(3, \infty)$.

2. (*Application*) Let $f(x) = x^2$ and $g(x) = x - 4$. Find $(f \circ g)(x)$ and state whether g is invertible; if so, find $g^{-1}(x)$.

Answer: $(f \circ g)(x) = (x - 4)^2$; g is invertible (linear, one-to-one), $g^{-1}(x) = x + 4$.

1.2 2.1 Exponential Growth and Decay

Core Concepts

- Exponential functions b^x and the base e
- Continuous vs. discrete growth models
- Growth/decay constant k ; doubling time and half-life
- Preview of the differential equation $\frac{dy}{dt} = ky$

Learning Objectives

1. Explain the qualitative difference between linear and exponential growth.
2. Derive the half-life/doubling-time formula from a given decay/growth constant.
3. Apply exponential models to population-growth and radioactive-decay word problems.
4. Evaluate, from a data trend, whether an exponential growth or decay model is appropriate.

Example Questions

1. (*Recall*) A population grows according to $P(t) = P_0 e^{kt}$ with $k = 0.05/\text{year}$. What does k represent?
Answer: The continuous relative (per-capita) growth rate: the population grows at 5% per year, compounded continuously.
2. (*Application*) A bacterial culture doubles every 3 hours. If it starts at 200 cells, find k and predict the population after 10 hours.
Answer: $2 = e^{3k} \Rightarrow k = \frac{\ln 2}{3} \approx 0.231$; $P(10) = 200e^{0.231(10)} \approx 200e^{2.31} \approx 2009$ cells.

1.3 2.2 Sequences

Core Concepts

- Sequence notation a_n
- Recursive vs. explicit definitions
- Convergence and divergence of a sequence
- Informal limit of a sequence as $n \rightarrow \infty$
- Monotonic and bounded sequences

Learning Objectives

1. Define a sequence both recursively and explicitly.
2. Determine whether a simple sequence converges or diverges.
3. Compute the limit of a sequence using algebraic limit laws.
4. Relate a discrete sequence model to the continuous growth model of Section 2.1.

Example Questions

1. (*Recall*) Does the sequence $a_n = \frac{n}{n+1}$ converge? If so, to what value?
Answer: Yes; $a_n \rightarrow 1$ as $n \rightarrow \infty$.
2. (*Application*) A drug dose recursion is $a_{n+1} = 0.5a_n + 10$, $a_0 = 0$ (mg, after each dosing interval). Find a_1, a_2, a_3 and conjecture the long-run (steady-state) value.
Answer: $a_1 = 10$, $a_2 = 15$, $a_3 = 17.5$; steady state solves $L = 0.5L + 10 \Rightarrow L = 20$ mg.

2 Module B: Limits and Continuity (Weeks 3–4)

2.1 3.1 Limits

Core Concepts

- Intuitive limit as $x \rightarrow a$; one-sided limits
- Limit laws: sum, product, quotient
- Formal ε - δ definition (Sec. 3.6)
- Nonexistence of a limit: oscillation, jump, unboundedness

Learning Objectives

1. Estimate a limit numerically and graphically.
2. Compute limits algebraically using limit laws.
3. Prove a simple limit statement using the formal ε - δ definition.
4. Diagnose why a limit fails to exist at a given point.

Why This Topic Is Historically Difficult

The ε - δ definition asks students to reason about two interdependent “smallness” quantities inside nested quantifiers ($\forall \varepsilon \exists \delta$), which is a genuine jump from the procedural algebra they have used up to this point. The notation itself (two Greek letters, an implication, an absolute-value inequality) is unfamiliar, so many students’ difficulty is more about parsing the symbols than about the underlying idea.

Mitigation: Introduce the idea in “tolerance/target” language before any symbols (“how close does x need to be to a so that $f(x)$ lands within a given target window?”), use a numerical guess-and-check table (pick ε , find a working δ) before asking for a general proof, and restrict formal proofs to linear functions only in this course. **Example Questions**

1. (*Recall*) Evaluate $\lim_{x \rightarrow 2} (3x^2 - x)$.

Answer: $3(4) - 2 = 10$.

2. (*Application*) Explain why $\lim_{x \rightarrow 0} \frac{|x|}{x}$ does not exist.

Answer: The left-hand limit is -1 and the right-hand limit is 1 ; since the one-sided limits differ, the two-sided limit does not exist.

3. (*Analysis*) Using the formal definition, show that $\lim_{x \rightarrow 3} (2x + 1) = 7$.

Rubric: Full credit requires: given $\varepsilon > 0$, choosing $\delta = \varepsilon/2$, and correctly showing $0 < |x - 3| < \delta \Rightarrow |(2x + 1) - 7| = 2|x - 3| < 2\delta = \varepsilon$.

2.2 3.2 Continuity

Core Concepts

- Three-part definition of continuity at a point
- Removable, jump, and infinite discontinuities
- Continuity on an interval
- Intermediate Value Theorem (IVT)

Learning Objectives

1. Test whether a function is continuous at a point using the three-part definition.
2. Classify a discontinuity by type.
3. Apply the IVT to justify the existence of a root or solution.
4. Construct a piecewise function exhibiting a specified type of discontinuity.

Example Questions

1. (*Recall*) Classify the discontinuity of $f(x) = \frac{x^2 - 4}{x - 2}$ at $x = 2$.

Answer: Removable (the factor $x - 2$ cancels; the limit exists but $f(2)$ is undefined).

2. (*Application*) Show that $f(x) = x^3 - x - 1$ has a root in $(1, 2)$.

Answer: f is continuous (polynomial); $f(1) = -1 < 0$ and $f(2) = 5 > 0$; by the IVT there exists $c \in (1, 2)$ with $f(c) = 0$.

2.3 3.3 Limits at Infinity

Core Concepts

- Horizontal asymptotes
- End behavior of rational, exponential, and logarithmic functions
- Formal ε - M definition (Sec. 3.6)
- Distinguishing limits at infinity from infinite limits

Learning Objectives

1. Compute limits as $x \rightarrow \pm\infty$ for rational and transcendental functions.
2. Identify horizontal asymptotes from a limit computation.
3. Distinguish, in words and notation, an infinite limit from a limit at infinity.

Example Questions

1. (*Recall*) Find $\lim_{x \rightarrow \infty} \frac{3x^2 + 1}{5x^2 - x}$.

Answer: $3/5$ (divide numerator and denominator by x^2).

2. (*Application*) Find all horizontal asymptotes of $f(x) = \frac{2x + 1}{\sqrt{x^2 + 1}}$.

Answer: $y = 2$ as $x \rightarrow \infty$ and $y = -2$ as $x \rightarrow -\infty$ (the sign of x matters once $\sqrt{x^2}$ is rewritten as $|x|$).

2.4 3.4 Trigonometric Limits and the Sandwich Theorem

Core Concepts

- $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$ and related limits
- Statement of the Sandwich (Squeeze) Theorem
- Bounding a function between two known functions

Learning Objectives

1. State and apply the Sandwich Theorem.
2. Derive $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$ using a geometric bounding argument.
3. Apply trigonometric limit results in later derivative derivations.

Example Questions

1. (*Recall*) Evaluate $\lim_{x \rightarrow 0} \frac{\sin(5x)}{x}$.

Answer: 5 (rewrite as $5 \cdot \frac{\sin 5x}{5x}$).

2. (*Application*) Given $-x^2 \leq g(x) \leq x^2$ for all x , find $\lim_{x \rightarrow 0} g(x)$ and justify using the Sandwich Theorem.

Answer: 0 , since both bounding functions $\rightarrow 0$ as $x \rightarrow 0$.

2.5 3.5 Properties of Continuous Functions

Core Concepts

- Preview of the Extreme Value Theorem

- Boundedness of a continuous function on a closed interval
- Continuity of compositions and algebraic combinations of continuous functions

Learning Objectives

1. Explain why a continuous function on $[a, b]$ must attain a maximum and minimum (preview of 5.1).
2. Prove continuity of a composite or combined function using the relevant theorems.
3. Apply these properties to simplify continuity arguments.

Example Questions

1. (*Recall*) Why is $h(x) = \sin(x^2 + 1)$ continuous everywhere?
Answer: It is a composition of continuous functions ($x^2 + 1$ is continuous, $\sin$ is continuous), and compositions of continuous functions are continuous.
2. (*Application*) f is continuous on $[0, 4]$ with $f(0) = -2$ and $f(4) = 6$. Must f attain the value 2 somewhere in $[0, 4]$? Justify.
Answer: Yes, by the IVT, since 2 lies between $f(0)$ and $f(4)$ and f is continuous on the interval.

3 Module C: Derivatives (Weeks 5–7)

3.1 4.1 Formal Definition of the Derivative

Core Concepts

- Difference quotient $\frac{f(x+h) - f(x)}{h}$
- Derivative as the limit of the difference quotient
- Derivative as instantaneous rate of change / slope of the tangent line
- Differentiability implies continuity (converse fails)

Learning Objectives

1. Compute a derivative directly from the limit definition for simple functions.
2. Explain the geometric and physical meaning of $f'(a)$.
3. Exhibit a function that is continuous but not differentiable at a point.

Example Questions

1. (*Recall*) Use the limit definition to find $f'(x)$ for $f(x) = x^2$.
Answer: $f'(x) = \lim_{h \rightarrow 0} \frac{(x+h)^2 - x^2}{h} = \lim_{h \rightarrow 0} (2x + h) = 2x$.
2. (*Application*) Show that $f(x) = |x|$ is continuous at $x = 0$ but not differentiable there.
Rubric: Full credit requires showing $\lim_{x \rightarrow 0} |x| = 0 = f(0)$ (continuity), and computing the left-hand difference quotient limit (-1) and right-hand limit (1) to show they disagree (non-differentiability).

3.2 4.2 Basic Rules of Derivatives

Core Concepts

- Constant rule; sum/difference rule; constant-multiple rule
- Linearity of differentiation

Learning Objectives

1. Apply the constant, sum, and constant-multiple rules to differentiate polynomials.
2. Justify these rules from the limit definition.

Example Questions

1. (*Recall*) Differentiate $f(x) = 5x^3 - 2x + 7$.
Answer: $f'(x) = 15x^2 - 2$.
2. (*Application*) Prove the constant-multiple rule $\frac{d}{dx}[cf(x)] = cf'(x)$ from the limit definition.
Rubric: Credit for correctly factoring c out of the difference quotient and applying the limit-of-a-constant-times-a-function law.

3.3 4.3 Power Rule and the Rules of Differentiation

Core Concepts

- Power rule for integer, rational, and negative exponents
- Combining the power rule with the basic rules

Learning Objectives

1. Apply the power rule to functions with rational and negative exponents.
2. Differentiate polynomial and radical functions fluently.

Example Questions

1. (*Recall*) Differentiate $f(x) = x^5$ and $g(x) = x^{-3}$.
Answer: $f'(x) = 5x^4$, $g'(x) = -3x^{-4}$.
2. (*Application*) Differentiate $f(x) = 3\sqrt{x} - \frac{4}{x^2}$.
Answer: Rewrite as $3x^{1/2} - 4x^{-2}$; $f'(x) = \frac{3}{2}x^{-1/2} + 8x^{-3}$.

3.4 4.4 Product and Quotient Rules

Core Concepts

- Product rule formula and its derivation
- Quotient rule formula and its derivation
- Common sign errors in the quotient rule

Learning Objectives

1. Derive the product rule from the limit definition.
2. Apply the product and quotient rules to composite algebraic expressions.
3. Diagnose and correct a quotient-rule sign error in sample work.

Example Questions

1. (*Recall*) Differentiate $f(x) = x^2 \sin x$.
Answer: $f'(x) = 2x \sin x + x^2 \cos x$.
2. (*Application*) A student writes $\frac{d}{dx} \left[\frac{x^2}{x+1} \right] = \frac{2x(x+1)+x^2}{(x+1)^2}$. Identify and correct the error.
Answer: The sign in the numerator is wrong; the correct quotient rule gives $\frac{2x(x+1)-x^2}{(x+1)^2} = \frac{x^2+2x}{(x+1)^2}$.

3.5 4.5 The Chain Rule

Core Concepts

- Composite functions $f(g(x))$
- Chain rule formula; “outside–inside” heuristic
- Chain rule with multiply-nested compositions

Learning Objectives

1. Identify the inner and outer functions in a composite expression.
2. Apply the chain rule to single and multiply-nested compositions.
3. Combine the chain rule with the product and quotient rules.

Why This Topic Is Historically Difficult

The most common error is not a conceptual gap but an execution gap: students differentiate the outer function correctly but forget to multiply by the derivative of the inner function, or misidentify which piece is “outside.” The error compounds silently when compositions are nested two or three layers deep, since each layer adds another factor that is easy to drop.

Mitigation: Teach an explicit “peel the onion, layer by layer” procedure and sequence worked examples from one layer of nesting to three. Use error-spotting exercises built from real (anonymized) student mistakes so learners practice catching the missing-factor error rather than only producing correct answers. **Example Questions**

1. (*Recall*) Differentiate $f(x) = (3x + 1)^5$.
Answer: $f'(x) = 15(3x + 1)^4$.
2. (*Application*) Differentiate $f(x) = \sin(\cos(x^2))$.
Answer: $f'(x) = \cos(\cos(x^2)) \cdot (-\sin(x^2)) \cdot 2x = -2x \sin(x^2) \cos(\cos(x^2))$.
3. (*Analysis*) A student differentiates $f(x) = e^{3x^2}$ and gets $f'(x) = e^{3x^2}$. Identify the missing step.
Answer: They forgot to multiply by the derivative of the inner function $3x^2$, namely $6x$; correct answer is $f'(x) = 6xe^{3x^2}$.

3.6 4.6 Implicit Functions and Implicit Differentiation

Core Concepts

- Implicitly defined curves (e.g., circles)
- Differentiating both sides with respect to x , treating y as a function of x
- Solving for dy/dx

Learning Objectives

1. Differentiate an implicitly defined relation.
2. Solve for dy/dx and evaluate the slope at a given point.
3. Apply implicit differentiation to a related-rates-style biological model.

Why This Topic Is Historically Difficult

Writing $\frac{d}{dx}(y^2) = 2y \frac{dy}{dx}$ requires silently treating y as $y(x)$ and applying the chain rule to a variable that has no explicit formula – a notational leap that differs from every derivative students have taken so far, where the “inside function” was always written out.

Mitigation: Before differentiating, have students explicitly rewrite y as $y(x)$ once on the page, then apply a fixed template: differentiate term by term, attach $\frac{dy}{dx}$ to every term containing y , then collect and solve. **Example Questions**

1. (*Recall*) Find dy/dx for $x^2 + y^2 = 25$.
Answer: $2x + 2y \frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx} = -\frac{x}{y}$.
2. (*Application*) Find the slope of the tangent line to $xy^2 + x^2 = 6$ at the point $(2, 1)$.
Answer: $y^2 + 2xy \frac{dy}{dx} + 2x = 0 \Rightarrow \frac{dy}{dx} = -\frac{y^2 + 2x}{2xy}$; at $(2, 1)$: $-\frac{1+4}{4} = -\frac{5}{4}$.

3.7 4.7 Higher Derivatives

Core Concepts

- Second- and higher-order derivative notation

- Physical interpretation (acceleration) and preview of concavity

Learning Objectives

1. Compute second and third derivatives of standard functions.
2. Interpret f'' in terms of concavity or acceleration.

Example Questions

1. (*Recall*) Find $f''(x)$ for $f(x) = x^4 - 3x^2$.
Answer: $f'(x) = 4x^3 - 6x$; $f''(x) = 12x^2 - 6$.
2. (*Application*) A particle's position is $s(t) = t^3 - 6t^2 + 9t$. Find its acceleration at $t = 2$ and state whether it is speeding up or slowing down at that instant (given $s'(2)$).
Answer: $s''(t) = 6t - 12$, so $s''(2) = 0$; with $s'(2) = 3(4) - 24 + 9 = -3 \neq 0$, acceleration is momentarily zero while velocity is nonzero, so the particle is neither speeding up nor slowing down at that exact instant.

3.8 4.8 Derivative of Trigonometric Functions

Core Concepts

- $\frac{d}{dx} \sin x = \cos x$, $\frac{d}{dx} \cos x = -\sin x$, derived from the limit definition and the trig limits of 3.4
- Derivatives of $\tan x$, $\cot x$, $\sec x$, $\csc x$

Learning Objectives

1. Derive $\frac{d}{dx} \sin x$ using the limit definition together with the trigonometric limits from Section 3.4.
2. Apply trig derivative rules together with the chain and product rules.

Example Questions

1. (*Recall*) Differentiate $f(x) = \tan x$.
Answer: $f'(x) = \sec^2 x$.
2. (*Application*) Differentiate $f(x) = x^2 \sec(3x)$.
Answer: $f'(x) = 2x \sec(3x) + x^2 \cdot 3 \sec(3x) \tan(3x) = 2x \sec(3x) + 3x^2 \sec(3x) \tan(3x)$.

3.9 4.9 Derivatives of Exponential Functions

Core Concepts

- $\frac{d}{dx} e^x = e^x$; $\frac{d}{dx} b^x = b^x \ln b$
- Definition of e as $\lim_{n \rightarrow \infty} (1 + 1/n)^n$, linking back to Sections 2.1/2.2

Learning Objectives

1. Derive $\frac{d}{dx} e^x = e^x$ from the limit definition.
2. Differentiate exponential functions of arbitrary base.
3. Apply exponential derivatives to rate-of-change problems in growth/decay contexts.

Example Questions

1. (*Recall*) Differentiate $f(x) = 3^x$.
Answer: $f'(x) = 3^x \ln 3$.
2. (*Application*) A population is modeled by $P(t) = 500e^{0.08t}$. Find the instantaneous growth rate $P'(t)$ and evaluate it at $t = 10$.
Answer: $P'(t) = 40e^{0.08t}$; $P'(10) = 40e^{0.8} \approx 89.0$ individuals per unit time.

3.10 4.10 Derivatives of Inverse Functions

Core Concepts

- General inverse function derivative formula $(f^{-1})'(x) = \frac{1}{f'(f^{-1}(x))}$
- Derivatives of $\ln x$, $\arcsin x$, $\arctan x$, etc.
- Logarithmic differentiation

Learning Objectives

1. Derive the general inverse-function derivative formula.
2. Differentiate logarithmic and inverse trigonometric functions.
3. Apply logarithmic differentiation to functions with variable exponents.

Example Questions

1. (*Recall*) Differentiate $f(x) = \ln(5x)$.
Answer: $f'(x) = \frac{1}{x}$.
2. (*Application*) Use logarithmic differentiation to differentiate $f(x) = x^x$ ($x > 0$).
Answer: $\ln f = x \ln x \Rightarrow \frac{f'}{f} = \ln x + 1 \Rightarrow f'(x) = x^x(\ln x + 1)$.

3.11 4.11 Linear Approximation

Core Concepts

- Tangent-line approximation $L(x) = f(a) + f'(a)(x - a)$
- Differentials $dy = f'(x) dx$
- Informal error estimation

Learning Objectives

1. Construct the linear approximation of a function at a point.
2. Use a linear approximation to estimate a function value near a .
3. Judge when a linear approximation is reliable versus unreliable.

Example Questions

1. (*Recall*) Find the linear approximation of $f(x) = \sqrt{x}$ at $a = 25$.
Answer: $f(25) = 5$, $f'(x) = \frac{1}{2\sqrt{x}}$, $f'(25) = \frac{1}{10}$; $L(x) = 5 + \frac{1}{10}(x - 25)$.
2. (*Application*) Use the result above to estimate $\sqrt{26.5}$, and discuss whether the estimate is likely an over- or under-estimate.
Answer: $L(26.5) = 5 + 0.15 = 5.15$; since $\sqrt{x}$ is concave down, the tangent line lies above the curve, so 5.15 is a slight overestimate of the true value (≈ 5.148).

4 Module D: Applications of Differentiation (Weeks 8–9)

4.1 5.1 Extreme and Mean Value Theorems

Core Concepts

- Absolute vs. local extrema
- Extreme Value Theorem (EVT)
- Fermat's Theorem and critical points
- Rolle's Theorem and the Mean Value Theorem (MVT)

Learning Objectives

1. State the hypotheses and conclusions of the EVT and the MVT.
2. Find the critical points of a function.
3. Apply the MVT to justify a statement about a function's behavior.

Example Questions

1. (*Recall*) Find the critical points of $f(x) = x^3 - 3x + 1$.

Answer: $f'(x) = 3x^2 - 3 = 0 \Rightarrow x = \pm 1$.

2. (*Application*) Show that $f(x) = x^3 + x - 1$ has exactly one real root using Rolle's Theorem (by contradiction) together with the IVT.

Rubric: Credit for (i) using IVT to show a root exists (e.g., $f(0) = -1 < 0 < f(1) = 1$), and (ii) arguing that $f'(x) = 3x^2 + 1 > 0$ always, so f is strictly increasing and cannot have two roots (if it did, Rolle's Theorem would require $f' = 0$ somewhere between them, a contradiction).

4.2 5.2 Monotonicity and Concavity

Core Concepts

- First derivative test for increasing/decreasing behavior
- Second derivative and concavity
- Sign charts

Learning Objectives

1. Determine intervals of increase/decrease using f' .
2. Determine concavity using f'' .
3. Construct a sign chart summarizing a function's behavior.

Example Questions

1. (*Recall*) Find the intervals where $f(x) = x^3 - 3x^2$ is increasing.

Answer: $f'(x) = 3x^2 - 6x = 3x(x - 2)$; increasing on $(-\infty, 0) \cup (2, \infty)$.

2. (*Application*) Determine the concavity of $f(x) = x^4 - 6x^2$ on all intervals.

Answer: $f''(x) = 12x^2 - 12 = 12(x - 1)(x + 1)$; concave up on $(-\infty, -1) \cup (1, \infty)$, concave down on $(-1, 1)$.

4.3 5.3 Extrema and Inflection Points

Core Concepts

- Classifying critical points via the first- or second-derivative test
- Inflection points (concavity change)

Learning Objectives

1. Classify critical points as local maxima, minima, or neither.
2. Locate inflection points and justify with a concavity-change argument.
3. Sketch a curve using first- and second-derivative information.

Why This Topic Is Historically Difficult

Students frequently conflate “critical point” with “extremum” (not every critical point is a max/min, e.g. $f(x) = x^3$ at $x = 0$) and conflate “ $f'' = 0$ ” with “inflection point” (concavity must actually *change* sign, e.g. $f(x) = x^4$ has $f''(0) = 0$ but no inflection point there). Both errors stem from treating a necessary condition as if it were sufficient.

Mitigation: Pair every rule with an explicit counterexample (x^3 for critical-but-not-extreme, x^4 for $f'' = 0$ -but-not-inflection) at first introduction, and require a sign-change check, not just a zero, before students may label a point an inflection point. **Example Questions**

1. (*Recall*) Classify the critical point of $f(x) = x^3$ at $x = 0$.

Answer: Neither a max nor a min ($f'(x) = 3x^2 \geq 0$ everywhere, so f never decreases).

2. (*Application*) Find and classify all local extrema and inflection points of $f(x) = x^4 - 4x^3$.
Answer: $f'(x) = 4x^3 - 12x^2 = 4x^2(x - 3)$, critical points $x = 0, 3$; sign analysis shows f decreasing on $(-\infty, 3)$, increasing on $(3, \infty)$, so $x = 3$ is a local min, $x = 0$ is neither.
 $f''(x) = 12x^2 - 24x = 12x(x - 2)$, sign changes at $x = 0$ and $x = 2$, so both are inflection points.

4.4 5.4 Optimization

Core Concepts

- Translating a word problem into an objective function
- Identifying constraints and the domain of the objective function
- Solving via critical points plus endpoint/boundary checks

Learning Objectives

1. Translate a real-world (biological or social-science) scenario into an optimization function.
2. Apply calculus techniques to find the optimal value.
3. Justify, via the first- or second-derivative test, that a critical point is the required max or min.

Why This Topic Is Historically Difficult

The bottleneck is rarely the calculus itself; it is the translation step. Turning prose into a labeled diagram, a constraint equation, and a single-variable objective function is a distinct skill from differentiating, and many students who can optimize a given function cannot construct one from a word problem. The cognitive load of doing both steps at once overwhelms working memory.

Mitigation: Teach a fixed five-step template (draw a diagram $\rightarrow$ label variables $\rightarrow$ write the constraint $\rightarrow$ substitute to get a one-variable objective $\rightarrow$ differentiate and classify), and sequence practice from simple geometric setups (rectangle with fixed perimeter) to biological ones (e.g., resource-allocation or population models) only after the template is automatic. **Example Questions**

1. (*Recall*) A rectangle has perimeter 40. Write its area as a function of one side length x .
Answer: $A(x) = x(20 - x)$.
2. (*Application*) Find the dimensions of the rectangle in the previous question that maximize area, and confirm it is a maximum.
Answer: $A'(x) = 20 - 2x = 0 \Rightarrow x = 10$; $A''(x) = -2 < 0$ confirms a maximum; the rectangle is a 10×10 square, area 100.
3. (*Analysis*) A culture's growth rate depends on nutrient concentration c via $R(c) = \frac{c}{1 + c^2}$. Find the nutrient concentration that maximizes the growth rate.
Answer: $R'(c) = \frac{(1 + c^2) - c(2c)}{(1 + c^2)^2} = \frac{1 - c^2}{(1 + c^2)^2} = 0 \Rightarrow c = 1$ (taking $c > 0$); sign change from positive to negative confirms a maximum at $c = 1$.

4.5 5.5 L'Hopital's Rule

Core Concepts

- Indeterminate forms $0/0$ and ∞/∞
- Statement and hypotheses of L'Hopital's Rule
- Algebraically converting other indeterminate forms ($0 \cdot \infty$, $\infty - \infty$, 0^0 , etc.) into $0/0$ or ∞/∞

Learning Objectives

1. Identify when L'Hopital's Rule applies.

2. Apply L'Hopital's Rule, including repeated application.
3. Convert a non-standard indeterminate form into $0/0$ or ∞/∞ before applying the rule.

Why This Topic Is Historically Difficult

Students often apply L'Hopital's Rule to limits that are not actually indeterminate, or differentiate the entire fraction using the quotient rule instead of differentiating numerator and denominator separately – confusing this rule with ordinary differentiation of a quotient.

Mitigation: Require students to write the verification “form: $0/0$ ” or “form: ∞/∞ ” explicitly before applying the rule, and use a side-by-side common-error walkthrough contrasting correct L'Hopital application against the quotient-rule mistake. **Example Questions**

1. (*Recall*) Evaluate $\lim_{x \rightarrow 0} \frac{\sin x}{x}$ using L'Hopital's Rule.
Answer: Form is $0/0$; differentiate top and bottom: $\lim_{x \rightarrow 0} \frac{\cos x}{1} = 1$.
2. (*Application*) Evaluate $\lim_{x \rightarrow \infty} x e^{-x}$.
Answer: Rewrite as $\lim_{x \rightarrow \infty} \frac{x}{e^x}$, form ∞/∞ ; apply L'Hopital: $\lim_{x \rightarrow \infty} \frac{1}{e^x} = 0$.

5 Module E: Integration (Weeks 10–11)

5.1 5.10 Antiderivatives

Core Concepts

- Definition of an antiderivative / indefinite integral
- Basic antiderivative rules: power, exponential, trigonometric
- The constant of integration C
- Initial value problems

Learning Objectives

1. Compute antiderivatives of elementary functions.
2. Solve an initial value problem to find a particular antiderivative.
3. Explain why any two antiderivatives of the same function differ by a constant.

Example Questions

1. (*Recall*) Find the general antiderivative of $f(x) = 4x^3 + \sin x$.
Answer: $F(x) = x^4 - \cos x + C$.
2. (*Application*) Find $f(x)$ given $f'(x) = 6x^2 - 2$ and $f(1) = 5$.
Answer: $f(x) = 2x^3 - 2x + C$; $f(1) = 2 - 2 + C = 5 \Rightarrow C = 5$, so $f(x) = 2x^3 - 2x + 5$.

5.2 6.1 The Definite Integral

Core Concepts

- Riemann sums: left, right, midpoint
- The definite integral as a limit of Riemann sums
- Area interpretation (signed area)
- Properties of definite integrals: linearity, additivity over intervals

Learning Objectives

1. Approximate a definite integral using a Riemann sum.
2. Set up and interpret a definite integral as signed area.
3. Apply properties of definite integrals to simplify a computation.

Why This Topic Is Historically Difficult

The definite integral asks students to accept a limit of an ever-finer sum of an ever-larger number of rectangles – an abstraction jump layered on top of unfamiliar notation ($\sum$, Δx , sample points x_i^*) all introduced at once. Many students can compute a Riemann sum mechanically without understanding what the limiting process represents.

Mitigation: Use a visual or numerical scaffold (a spreadsheet or applet showing the sum converging as the number of rectangles increases) before introducing sigma notation formally, and separate “compute a sum for a fixed n ” practice from “interpret the limit as $n \rightarrow \infty$ ” practice. **Example Questions**

1. (*Recall*) Approximate $\int_0^2 x^2 dx$ using a right-endpoint Riemann sum with $n = 4$ rectangles.
Answer: $\Delta x = 0.5$; right endpoints 0.5, 1, 1.5, 2; sum = $0.5(0.25 + 1 + 2.25 + 4) = 3.75$.
2. (*Application*) Interpret $\int_{-2}^2 x dx = 0$ geometrically, and explain why this does *not* mean the region has zero area.
Answer: The integral gives *signed* area; the region above the axis on $[0, 2]$ and below the axis on $[-2, 0]$ have equal magnitude but opposite sign, so they cancel, even though the total (unsigned) area enclosed is nonzero.

5.3 6.2 The Fundamental Theorem of Calculus

Core Concepts

- FTC Part 1: the derivative of an accumulation function
- FTC Part 2: evaluating a definite integral via an antiderivative
- The connection between differentiation and integration

Learning Objectives

1. State both parts of the Fundamental Theorem of Calculus.
2. Apply FTC Part 2 to evaluate a definite integral.
3. Apply FTC Part 1 to differentiate a function defined by an integral.

Why This Topic Is Historically Difficult

Students often memorize FTC Part 2 as a computational shortcut without grasping *why* integration undoes differentiation, and they routinely confuse the two parts – particularly Part 1, which produces a derivative of an accumulation function rather than a number.

Mitigation: Present both parts in an explicit side-by-side comparison table with distinct worked examples for each, and frame Part 1 with a “rate in $\rightarrow$ total accumulated” narrative (e.g., a rate function whose accumulation gives total change) before introducing the symbolic statement $\frac{d}{dx} \int_a^x f(t) dt = f(x)$. **Example Questions**

1. (*Recall*) Evaluate $\int_1^3 (2x + 1) dx$ using FTC Part 2.
Answer: Antiderivative $x^2 + x$; $(9 + 3) - (1 + 1) = 10$.
2. (*Application*) Let $g(x) = \int_0^x \sqrt{t^2 + 1} dt$. Find $g'(x)$.
Answer: By FTC Part 1, $g'(x) = \sqrt{x^2 + 1}$.

5.4 6.3 Applications of Integration

Core Concepts

- Area between two curves

- Average value of a function on an interval
- Accumulated change (e.g., total change from a given rate function)

Learning Objectives

1. Set up and compute the area between two curves.
2. Compute the average value of a function on an interval.
3. Apply the integral to accumulate a rate into a total change in a biological or social-science context.

Example Questions

1. (*Recall*) Find the area between $y = x^2$ and $y = 4$ on the interval where $x^2 \leq 4$.
Answer: $\int_{-2}^2 (4 - x^2) dx = \left[4x - \frac{x^3}{3}\right]_{-2}^2 = \frac{32}{3}$.
2. (*Application*) A population's growth rate is $R(t) = 50e^{0.1t}$ individuals/year. Find the total population increase between $t = 0$ and $t = 5$.
Answer: $\int_0^5 50e^{0.1t} dt = 500(e^{0.5} - 1) \approx 500(0.6487) \approx 324.4$ individuals.

6 Module F: Applications of the Integral (Week 12)

6.1 7.1 The Substitution Rule

Core Concepts

- u -substitution for indefinite integrals
- Changing the limits of integration for a definite integral
- Recognizing composite-function integrands

Learning Objectives

1. Identify an appropriate substitution $u = g(x)$ for a given integral.
2. Apply substitution to evaluate indefinite and definite integrals.
3. Correctly adjust the limits of integration when substituting in a definite integral.

Why This Topic Is Historically Difficult

Substitution is essentially the chain rule run in reverse, so students who never fully internalized “spot the inner function” in 4.5 struggle equally here, now compounded by the extra bookkeeping step of also changing dx and (for definite integrals) the limits of integration – a step that is easy to forget entirely.

Mitigation: Run explicit “spot the inside function” drills on integrands before ever computing an antiderivative, and require a written limit-conversion table (x -limits $\rightarrow u$ -limits) as a mandatory step for every definite-integral substitution problem. **Example Questions**

1. (*Recall*) Evaluate $\int 2x(x^2 + 1)^4 dx$.

Answer: Let $u = x^2 + 1$, $du = 2x dx$; integral $= \int u^4 du = \frac{u^5}{5} + C = \frac{(x^2+1)^5}{5} + C$.

2. (*Application*) Evaluate $\int_0^2 xe^{x^2} dx$, changing the limits of integration explicitly.

Answer: $u = x^2$, $du = 2x dx$; when $x = 0$, $u = 0$; when $x = 2$, $u = 4$; integral $= \frac{1}{2} \int_0^4 e^u du = \frac{1}{2}(e^4 - 1)$.

6.2 7.2 Integration by Parts and Practicing Integration

Core Concepts

- Integration by parts formula, derived from the product rule

- LIATE/ILATE heuristic for choosing u and dv
- Repeated and tabular integration by parts
- General strategy selection among substitution, parts, and direct integration

Learning Objectives

1. Derive the integration-by-parts formula from the product rule.
2. Apply integration by parts, including cases requiring repetition.
3. Select the correct integration technique for a given mixed problem set.

Why This Topic Is Historically Difficult

A poor choice of u and dv produces a *harder* integral than the original, and students frequently apply the LIATE heuristic mechanically without understanding why it tends to work, so they cannot adapt when it fails or when repeated integration by parts is needed.

Mitigation: Show an explicit “wrong choice vs. right choice” worked comparison for the same integral, then build toward problems (e.g., $\int x^2 e^x dx$) that require recognizing the need for repetition or the tabular shortcut. **Example Questions**

1. (*Recall*) Evaluate $\int x e^x dx$.
Answer: $u = x, dv = e^x dx; = x e^x - \int e^x dx = x e^x - e^x + C$.
2. (*Application*) Evaluate $\int x^2 e^x dx$ (requires integration by parts twice).
Answer: $= x^2 e^x - 2x e^x + 2e^x + C$ (first pass: $u = x^2$; second pass on the resulting $\int 2x e^x dx$: $u = 2x$).
3. (*Analysis*) A student sets $u = e^x, dv = x dx$ for $\int x e^x dx$ and gets stuck. Explain why this choice is unproductive and what the better choice is.
Answer: This choice makes $v = \frac{x^2}{2}$, producing a *more* complicated integral $\int \frac{x^2}{2} e^x dx$; the better choice is $u = x$ (which differentiates to something simpler) and $dv = e^x dx$ (which is easy to integrate).

6.3 7.3 Rational Functions and Partial Fractions

Core Concepts

- Partial fraction decomposition for distinct linear factors (with brief mention of repeated and irreducible quadratic factors)
- Integrating each decomposed term
- Polynomial long division when the numerator’s degree is at least the denominator’s

Learning Objectives

1. Decompose a proper rational function into partial fractions.
2. Integrate each partial-fraction term.
3. Apply polynomial long division before decomposition when required.

Why This Topic Is Historically Difficult

This topic stacks three separate skills – factoring, solving a system of equations for unknown coefficients, and integration – into a single problem, and an algebra slip in the coefficient system silently invalidates the entire integral that follows, making errors hard for students to locate on their own.

Mitigation: Split practice into two stages: first, decomposition-only problems with no integration required, graded purely on the coefficient system; only after that skill is solid, combine decomposition with integration in full end-to-end problems. **Example Questions**

1. (*Recall*) Decompose $\frac{5x - 1}{(x - 1)(x + 2)}$ into partial fractions.

Answer: $\frac{5x-1}{(x-1)(x+2)} = \frac{A}{x-1} + \frac{B}{x+2}$; solving gives $A = \frac{4}{3}$, $B = \frac{11}{3}$.

2. (*Application*) Evaluate $\int \frac{5x-1}{(x-1)(x+2)} dx$ using the decomposition above.

Answer: $\frac{4}{3} \ln|x-1| + \frac{11}{3} \ln|x+2| + C$.

7 Prerequisite Map

Table 2: Topic-by-topic prerequisite dependencies

Topic	Prerequisite(s)	Where Taught
Ch. 1 Functions	Algebra, function notation	Assumed background (pre-calculus)
2.1 Exp. Growth/Decay	Exponent rules; function notation	Ch. 1 (this course); exponent rules assumed background
2.2 Sequences	Function notation; informal notion of a limit	Ch. 1 (this course); informal limit notion previewed here, formalized in 3.1
3.1 Limits	Function evaluation and algebra	Ch. 1
3.2 Continuity	Limits	3.1
3.3 Limits at Infinity	Limits; rational-function behavior; exponent rules	3.1; Ch. 1
3.4 Trig Limits & Sandwich Thm.	Limits; unit-circle trig values	3.1; Ch. 1
3.5 Properties of Cont. Functions	Continuity; limits	3.2; 3.1
4.1 Def. of Derivative	Limits; secant-line slope	3.1
4.2 Basic Rules	Derivative definition	4.1
4.3 Power Rule	Basic rules; exponent laws	4.2; Ch. 1
4.4 Product/Quotient Rules	Basic and power rules; derivative definition	4.2–4.3
4.5 Chain Rule	Function composition; product/quotient rules	Ch. 1; 4.2–4.4
4.6 Implicit Diff.	Chain rule	4.5
4.7 Higher Derivatives	Basic differentiation rules	4.2–4.5
4.8 Trig Derivatives	Trig limits; chain rule; trig identities	3.4; 4.5; Ch. 1
4.9 Exponential Derivatives	Definition of e ; chain rule	2.1–2.2; 4.5
4.10 Inverse Fn. Derivatives	Inverse functions; chain rule; implicit diff.	Ch. 1; 4.5; 4.6
4.11 Linear Approximation	Derivative as tangent slope	4.1
5.1 EVT/MVT	Continuity; differentiability	3.2; 4.1

Topic	Prerequisite(s)	Where Taught
5.2 Monotonicity/Concavity	First and second derivatives	4.1; 4.7
5.3 Extrema/Inflection Pts.	Monotonicity/concavity; critical points	5.2; 5.1
5.4 Optimization	Extrema; full differentiation toolkit	5.3; Module C
5.5 L'Hopital's Rule	Limits; differentiation toolkit	3.1, 3.3; Module C
5.10 Antiderivatives	All differentiation rules (reversed)	Module C
6.1 Definite Integral	Limits; area/geometry concepts	3.1
6.2 FTC	Antiderivatives; definite integral	5.10; 6.1
6.3 Applications of Integration	FTC	6.2
7.1 Substitution Rule	Chain rule (reversed); antiderivatives	4.5; 5.10
7.2 Integration by Parts	Product rule (reversed); substitution	4.4; 7.1
7.3 Partial Fractions	Factoring/systems of equations; polynomial division; log antiderivative	Ch. 1 (background); 4.10; 5.10

8 Assessment Plan

Table 3: Assessment schedule across the 12-week term

Module	Assessment	Timing	Rationale
A	Diagnostic quiz + Homework Set 1	End of Wk 2	Low-stakes diagnostic surfaces pre-calculus gaps (domain/range, exponent rules) before they compound in limits and derivatives.
B	In-class ε - δ exercise	Wk 3	Surfaces the quantifier/notation misconception immediately, before continuity content builds on an unstable grasp of limits.
B	Quiz on limits & continuity + Homework Set 2	End of Wk 4	Checks limit-law fluency and continuity classification before differentiation begins.

Module	Assessment	Timing	Rationale
C	In-class chain-rule error-spotting exercise	Wk 6	Catches the missing-inner-derivative error (the most common chain-rule mistake) while it is still cheap to fix, before it propagates into implicit differentiation and later integration by substitution.
C	Midterm 1 (limits through derivatives) + Homework Sets 3–4	End of Wk 7	Assesses cumulative differentiation fluency at a natural break point before moving to applications.
D	In-class optimization workshop	Wk 8	Scaffolds the word-problem-to-objective-function translation skill in a low-stakes, collaborative setting before it is tested individually.
D	Quiz on L'Hopital's Rule + Homework Set 5	Wk 9	Targets indeterminate-form misidentification and the quotient-rule confusion before the final exam.
E	Homework Set 6	Wk 10	Reinforces Riemann-sum and antiderivative mechanics while FTC is being introduced.
E	Midterm 2 (applications of differentiation through FTC)	End of Wk 11	Checks integration fundamentals at a second natural break point, since Module F techniques depend on them.
F	Homework Set 7 (integration techniques)	Wk 12	Verifies fluency in substitution, parts, and partial fractions, and in choosing among them.
F	Comprehensive Final Exam	Finals period	Assesses cumulative mastery across all six modules.
F	Optional applied project	Due finals period	Lets students (Bio. & Soc. Sci. audience) apply differentiation/integration to a real dataset or model, reinforcing course relevance.